NovaTech Solutions Pvt. Ltd.

Database Backup Manual

Document ID: TECH-DB-001

Version: 2.0

Effective Date: January 2026

Prepared By: Platform Engineering & Database Administration Team

Classification: Internal Use Only

Document Control

Field	Value
Document	Database Backup Manual
Database Platform	PostgreSQL 15
Version	2.0
Owner	Database Administration Team
Review Cycle	Every 6 Months
Last Updated	January 2026

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1. Introduction

This manual defines the backup policies and operational procedures for all production, staging, and development databases used by the NovaDesk Enterprise Service Management Platform.

The objectives of this document are to:

- Protect business-critical data.
- Support disaster recovery.
- Ensure business continuity.
- Minimize data loss.
- Meet internal compliance requirements.
- Standardize backup operations.

This document applies to:

- Production PostgreSQL databases
- Staging databases
- Development databases
- Reporting databases

Database backup activities are performed by authorized Database Administrators (DBAs) or approved automated systems.

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2. Backup Objectives

The NovaTech backup strategy is designed around two key objectives.

Recovery Point Objective (RPO)

The maximum acceptable amount of data loss.

Environment	RPO
Production	15 Minutes
Staging	4 Hours
Development	24 Hours

Continuous archive logging is enabled for production systems to support point-in-time recovery.

Recovery Time Objective (RTO)

The maximum acceptable time required to restore database services.

Environment	RTO
Production	2 Hours
Staging	4 Hours
Development	8 Hours

Operational procedures are designed to meet these recovery targets under normal conditions.

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3. Backup Strategy

NovaDesk follows a multi-layered backup strategy combining full, incremental, and transaction log backups.

The strategy is based on the **3-2-1 backup principle**:

- Maintain at least **3 copies** of critical data.
- Store backups on **2 different storage media**.
- Keep at least **1 copy off-site**.

This approach reduces the risk of data loss due to hardware failure, accidental deletion, ransomware, or site outages.

Backup Scope

The following data is included in scheduled backups:

- Application databases
- User accounts
- Configuration tables
- Workflow definitions
- Audit logs
- Knowledge Base content
- Asset inventory
- Ticket history
- Stored procedures
- Database roles

The following are excluded:

- Temporary tables

- Cache data
- Session data
- Application binaries
- Operating system files

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4. Backup Types

NovaDesk uses several backup types to balance recovery capability and storage efficiency.

Full Backup

A full backup contains a complete copy of the database.

Characteristics:

- Largest backup size
- Longest backup duration
- Simplest restoration process

Full backups serve as the baseline for incremental recovery.

Incremental Backup

Incremental backups contain only data changed since the previous backup.

Benefits:

- Reduced storage requirements
- Faster backup execution
- Efficient daily protection

Incremental backups require the most recent full backup during restoration.

Differential Backup

Differential backups contain all changes since the last full backup.

Compared with incremental backups:

- Larger backup size
- Faster restoration
- Simpler recovery chain

Differential backups are retained for operational recovery scenarios.

Transaction Log Backup

Production databases use transaction log backups to support point-in-time recovery.

Transaction log backups:

- Capture committed transactions
- Reduce potential data loss
- Enable recovery to a specific timestamp

Transaction log backups are generated every **15 minutes** for production systems.

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5. Backup Schedule

The following schedule applies to standard production environments.

Backup Type	Frequency	Time
Full Backup	Weekly	Sunday 02:00
Incremental Backup	Daily	02:00
Transaction Log Backup	Every 15 Minutes	Continuous
Differential Backup	Wednesday & Friday	02:00

Backup windows are selected to minimize impact on production workloads.

Staging Environment

Backup Type	Frequency
Full Backup	Weekly
Incremental Backup	Daily

Transaction log backups are not enabled by default for staging systems.

Development Environment

Development databases receive:

- Weekly Full Backup
- Daily Incremental Backup

Developers should not rely on development backups for long-term data preservation.

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6. Backup Retention Policy

Backup retention periods vary according to environment and regulatory requirements.

Production

Backup	Retention
Full Backup	90 Days
Incremental Backup	30 Days
Differential Backup	30 Days
Transaction Logs	14 Days

Staging

Backup	Retention
Full Backup	30 Days
Incremental Backup	14 Days

Development

Backup	Retention
Full Backup	14 Days
Incremental Backup	7 Days

Expired backups are automatically removed according to the retention policy after verification that newer backups are available.

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7. Backup Storage Locations

To improve resilience, backups are stored in multiple secure locations.

Primary Storage

Production backups are stored on dedicated backup storage separate from the production database servers.

Secondary Storage

A replicated copy is maintained within a secondary data center to support infrastructure failures.

Off-Site Storage

Encrypted backup copies are transferred daily to an approved off-site cloud storage repository.

Off-site storage protects against catastrophic events affecting the primary data center.

Storage Requirements

Backup storage systems must provide:

- Redundant storage
- Integrity verification
- Encryption at rest
- Access logging
- Automatic capacity monitoring

Storage utilization is reviewed weekly by the Database Administration Team.

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8. Backup Naming Convention

Consistent naming simplifies backup identification and restoration.

Standard format:

<Environment>_<DatabaseName>_<BackupType>_<YYYYMMDD_HHMMSS>.bak

Example:

PROD_NovaDesk_FULL_20260118_020000.bak

Incremental example:

PROD_NovaDesk_INC_20260119_020000.bak

Transaction log example:

PROD_NovaDesk_LOG_20260119_021500.trn

File names must not contain spaces or special characters other than underscores.

Backup Metadata

Each backup operation records:

- Backup ID
- Database Name
- Backup Type
- Start Time
- End Time
- Duration
- Backup Size
- Compression Status
- Encryption Status
- Operator
- Verification Status

This metadata is stored for audit and operational reporting.

Revision History

Version	Date	Description
1.0	January 2024	Initial Backup Manual
1.5	July 2025	Added Differential Backup Strategy
2.0	January 2026	Updated RPO, RTO, Retention Policy, and Storage Standards

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10. PostgreSQL Backup Procedures

NovaDesk uses PostgreSQL 15 as its primary database platform.

Database backups are executed using approved backup scripts that internally invoke PostgreSQL utilities such as pg_dump, pg_restore, and transaction log archiving mechanisms.

Manual backup execution is restricted to authorized Database Administrators.

Full Backup Procedure

Before initiating a full backup:

- Verify sufficient storage capacity.
- Confirm database health.
- Ensure no maintenance activities are in progress.
- Verify backup destination availability.

Backup workflow:

1. Validate database connectivity.
2. Initiate full backup.
3. Compress backup files.
4. Encrypt backup package.
5. Calculate checksum.
6. Store backup in the primary repository.
7. Replicate backup to secondary storage.
8. Copy encrypted backup to off-site storage.
9. Record backup metadata.
10. Perform backup verification.

Incremental Backup Procedure

Incremental backups capture only changes since the previous backup.

Procedure:

1. Validate previous backup completion.
2. Capture changed data.
3. Compress backup.
4. Encrypt backup.
5. Replicate backup.
6. Record execution details.

Incremental backups depend on the latest successful full backup.

Transaction Log Archiving

Production databases archive transaction logs continuously.

Archive operations include:

- Validate archive destination
- Copy completed transaction logs
- Verify successful transfer
- Update archive catalog

Missing transaction logs may prevent successful point-in-time recovery.

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11. Database Restore Procedures

Database restoration should only be performed after approval from the Database Administration Team and the designated Incident Manager for production systems.

Restore Preparation Checklist

Before restoring:

- Identify the recovery objective.
- Verify backup availability.
- Confirm backup integrity.
- Validate target environment.
- Notify stakeholders.
- Confirm rollback strategy.

Full Database Restore

General restore workflow:

1. Stop application services.
2. Prevent new database connections.
3. Restore the latest full backup.
4. Restore required incremental or differential backups.
5. Apply transaction logs if point-in-time recovery is required.
6. Validate restored database.
7. Start application services.
8. Execute smoke tests.
9. Monitor application health.

Point-in-Time Recovery

Point-in-time recovery (PITR) allows restoration to a specific timestamp using archived transaction logs.

Typical use cases include:

- Accidental record deletion
- User errors
- Data corruption
- Partial system failures

Recovery should target the last known consistent transaction before the incident.

Restore Validation

After every restore:

Verify:

- Database starts successfully.
- All expected schemas exist.
- Table counts are reasonable.
- User authentication functions correctly.
- Application connectivity is restored.
- Recent transactions are available.
- Reports execute successfully.

Restores are not considered complete until validation succeeds.

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12. Backup Verification

Every backup must be verified before being considered usable.

Verification Objectives

Verification confirms:

- Backup completed successfully.
- File is readable.
- Encryption succeeded.

- Compression succeeded.
- Checksum matches.
- File size is reasonable.
- Backup metadata is complete.

Checksum Validation

Every backup package includes a checksum.

Verification compares:

- Calculated checksum
- Stored checksum

Checksum mismatches indicate potential corruption and require immediate investigation.

Restore Testing

Verification should include periodic restoration testing.

Recommended schedule:

Environment	Restore Test Frequency
Production	Monthly
Staging	Monthly
Development	Quarterly

A backup is not considered reliable until restoration has been successfully tested.

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13. Disaster Recovery Procedures

Disaster Recovery (DR) restores business operations after major failures.

Disaster Scenarios

Examples include:

- Storage failure
- Database corruption
- Ransomware attack
- Data center outage

- Hardware failure
- Human error
- Power outage

Disaster Recovery Workflow

1. Declare incident.
2. Assemble response team.
3. Assess impact.
4. Select recovery strategy.
5. Restore infrastructure.
6. Restore database.
7. Validate applications.
8. Resume business operations.
9. Conduct post-incident review.

Recovery Priorities

Recovery order:

1. Authentication services
2. Production database
3. Ticket Management
4. Knowledge Base
5. Asset Management
6. Reporting services
7. Background processing

Critical business functions receive priority during recovery.

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14. Backup Automation

Backup operations are automated to reduce manual effort and improve reliability.

Automated Tasks

Scheduled automation performs:

- Backup initiation
- Compression
- Encryption
- Replication
- Integrity verification
- Retention cleanup
- Notification generation

Manual intervention is required only when exceptions occur.

Scheduled Jobs

Automated jobs execute according to the approved backup schedule defined in Part 1.

Failed jobs automatically generate alerts for the Database Administration Team.

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15. Monitoring

Backup operations are continuously monitored.

Monitored Metrics

The monitoring platform tracks:

- Backup duration
- Backup size
- Storage utilization
- Compression ratio
- Verification status
- Failed jobs
- Replication status
- Archive lag

Historical metrics support capacity planning and trend analysis.

Alert Conditions

Alerts are generated for:

- Backup failure
- Verification failure
- Storage capacity below 15%
- Replication failure
- Archive delay
- Excessive backup duration
- Missing scheduled backup

Critical alerts are delivered immediately to the on-call DBA.

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16. Logging

Every backup activity generates structured audit logs.

Logged Information

Backup logs include:

- Backup ID
- Database Name
- Backup Type
- Start Time
- End Time
- Duration
- File Size
- Compression Status
- Encryption Status
- Verification Result
- Operator
- Host Name
- Storage Destination

Logs are retained according to the organization's operational logging policy.

Audit Trail

Every restore operation records:

- Restore Request ID
- Approval Reference
- Operator
- Backup Used
- Recovery Time
- Validation Result
- Completion Status

Audit records support compliance reviews and post-incident investigations.

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17. Operational Best Practices

Database Administrators should:

- Monitor backup completion daily.
- Verify backup integrity after every execution.
- Test restores regularly.
- Maintain off-site copies.
- Review storage capacity weekly.
- Rotate encryption keys according to policy.
- Document all restore activities.
- Investigate failed backups immediately.
- Keep backup software updated.
- Review retention policies every six months.

Operational consistency significantly improves recovery reliability.

18. Backup Security

Database backups contain highly sensitive business information and must be protected throughout their lifecycle.

Backup security controls apply to:

- Production backups
- Staging backups

- Development backups
- Archive copies
- Off-site backup repositories

All backup operations must comply with NovaTech's Information Security Policy.

Security Objectives

Backup security aims to ensure:

- Confidentiality
- Integrity
- Availability
- Authenticity
- Traceability

Security controls should prevent unauthorized access, modification, deletion, or disclosure of backup data.

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19. Encryption

All production backup files must be encrypted before being transferred or stored.

Encryption Standards

NovaTech requires:

Component	Standard
Backup Encryption	AES-256
Data in Transit	TLS 1.2 or later
Integrity Verification	SHA-256 Checksum

Encryption keys must never be stored with backup files.

Key Management

Encryption keys are managed through the organization's approved Key Management System (KMS).

Requirements include:

- Annual key rotation
- Role-based access
- Audit logging
- Secure key backup
- Emergency recovery procedures

Only authorized security administrators may manage encryption keys.

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20. Access Control

Access to backup systems follows the principle of least privilege.

Authorized Roles

Role	Responsibilities
Database Administrator	Backup and restore operations
Platform Engineer	Infrastructure maintenance
Information Security	Audit and compliance reviews
Operations Manager	Operational approval and oversight

Backup repositories are not directly accessible to application users.

Authentication

Backup administration requires:

- Corporate identity authentication
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)

Privileged access is reviewed every six months.

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21. Compliance Requirements

Backup operations support organizational and regulatory compliance objectives.

Audit Requirements

Backup records must include:

- Backup identifier
- Database name
- Backup type
- Execution time
- Operator
- Verification result
- Storage location
- Encryption status

Audit records are retained for **seven years**.

Change Management

Changes to backup schedules, retention policies, or restore procedures require:

1. Documented change request.
2. Technical review.
3. Risk assessment.
4. Management approval.
5. Implementation during an approved maintenance window.

Emergency changes must be documented retrospectively.

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22. Backup Testing

Backup testing verifies that backup data can be restored successfully.

Testing Schedule

Activity	Frequency
Backup Verification	Every Backup
Restore Test	Monthly
Disaster Recovery Exercise	Quarterly
Full Recovery Simulation	Annually

Testing results should be documented and reviewed.

Test Validation

Every restore test should verify:

- Database startup
- Schema integrity
- Table accessibility
- User authentication
- Application connectivity
- Recent transactional data
- Reporting functionality

Any discrepancies should be investigated immediately.

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23. Troubleshooting Guide

The following table summarizes common backup-related issues.

Issue	Possible Cause	Recommended Resolution
Backup job failed	Insufficient storage	Increase available storage and rerun the job
Backup verification failed	File corruption	Recreate the backup and validate checksums
Restore failed	Missing transaction log	Verify archive completeness and repeat recovery
Backup duration increased	Database growth	Review storage performance and optimize schedules
Replication failed	Network interruption	Restore network connectivity and retry replication
Low storage warning	Capacity threshold reached	Archive or remove expired backups according to policy

Persistent issues should be escalated to the Database Administration Team.

Incident Escalation

Critical backup incidents include:

- Backup failure for production systems
- Backup corruption

- Loss of backup repository
- Failed disaster recovery exercise
- Encryption key compromise

Critical incidents require immediate notification of:

- On-call Database Administrator
- Platform Engineering Lead
- Information Security Team
- Operations Manager

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24. Maintenance Checklist

Database Administrators should complete the following routine activities.

Daily

- Verify backup completion.
- Review backup logs.
- Confirm replication success.
- Investigate failed jobs.

Weekly

- Review storage utilization.
- Verify backup retention cleanup.
- Check backup repository health.
- Validate monitoring alerts.

Monthly

- Perform restore testing.
- Review backup performance.
- Confirm encryption status.
- Validate backup automation jobs.

Quarterly

- Execute disaster recovery exercises.
- Review retention policies.
- Validate off-site backup accessibility.
- Review privileged access assignments.

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25. Operational Best Practices

NovaTech recommends the following operational practices:

- Keep multiple backup copies.
- Verify every backup.
- Test restores regularly.
- Encrypt all backup files.
- Separate backup infrastructure from production systems.
- Monitor backup health continuously.
- Document all restore operations.
- Maintain accurate operational runbooks.
- Automate routine tasks where appropriate.
- Periodically review recovery objectives.

Following these practices improves resilience and reduces recovery risk.

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26. Frequently Asked Questions

Q1. Which encryption standard is used for production backups?

Production backups are encrypted using **AES-256**.

Q2. How is backup integrity verified?

Integrity is verified using **SHA-256 checksums** before backups are accepted.

Q3. How often should disaster recovery exercises be performed?

Quarterly.

Q4. Who is authorized to perform production database restores?

Authorized Database Administrators following the approved restore process and required operational approvals.

Q5. How long are audit records retained?

Backup audit records are retained for **seven years**.

Q6. How frequently should privileged access be reviewed?

Every **six months**.

Q7. What should be done if backup verification fails?

Investigate the failure, recreate the backup, validate integrity, and document the incident.

Q8. What should be reviewed weekly?

Storage utilization, repository health, retention cleanup, and monitoring alerts.

Q9. Are backup repositories accessible to application users?

No. Access is restricted to authorized administrative roles.

Q10. What is the purpose of the annual recovery simulation?

To validate that complete business recovery objectives can be achieved under realistic disaster conditions.

Revision History

Version	Date	Description
1.0	January 2024	Initial Backup Manual
1.5	July 2025	Added Disaster Recovery Procedures
2.0	January 2026	Added Security, Compliance, Backup Testing, and Operational Governance

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Related Documents

- NovaDesk Developer Guide (TECH-DEV-001)
- NovaDesk REST API Documentation (TECH-API-001)
- Information Security Policy (COMP-SEC-003)
- Business Continuity Plan (OPS-BCP-002)
- Disaster Recovery Runbook (OPS-DR-001)